

Cognitive Stability of Chain-of-Thought Reasoning in LLMs Under Semantic Perturbation

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Abstract—Chain-of-thought (CoT) prompting improves final-answer accuracy on many reasoning tasks, yet benchmark accuracy alone does not establish whether reasoning behavior is cognitively stable. This paper develops a rigorous framework for testing the semantic stability of CoT trajectories under meaning-preserving perturbations. We formalize cognitive stability as invariance of intermediate reasoning under equivalent prompt reformulations and introduce the Cognitive Collapse Index (CCI), a trajectory-level metric that quantifies reasoning degradation independent of final-answer correctness. The paper contributes a perturbation taxonomy, a quality-controlled perturbation generation pipeline, model-agnostic scoring procedures, and a full statistical reporting protocol for cross-model comparisons. We further define complementary metrics for answer preservation, reasoning calibration, and perturbation sensitivity concentration. The framework is designed for contemporary LLM families and common reasoning benchmarks, including arithmetic, commonsense, and multi-hop QA tasks. Unlike evaluation pipelines that collapse all behavior into one accuracy value, the proposed protocol distinguishes output correctness from process robustness and surfaces decoupling cases where answers remain correct while reasoning collapses. This distinction is central to cognitive machine intelligence, trustworthy deployment, and human-AI collaboration in high-stakes decision settings.

Index Terms—cognitive machine intelligence, chain-of-thought, reasoning robustness, semantic perturbation, large language models, evaluation metrics, trustworthy AI

I. INTRODUCTION

Large language models (LLMs) have reached strong benchmark performance in reasoning-intensive tasks with chain-of-thought prompting [1]–[3]. However, most evaluation pipelines still score only the final answer [4]–[7]. This creates a blind spot: two systems can reach similar answer accuracy while exhibiting very different reasoning stability under semantically equivalent prompt variants.

From a cognitive perspective, robust reasoning should be relatively invariant to representational changes that preserve meaning. Humans usually maintain logical structure across

paraphrases, lexical substitutions, and abstraction shifts. By contrast, recent studies show that LLM behavior can be sensitive to prompt wording and format [8]–[10]. The same concern appears in faithfulness research, where generated rationales may not faithfully reflect latent decision mechanisms [11], [12]. These findings motivate trajectory-level evaluation, not only answer-level scoring.

This paper presents a complete cognitive-stability framework for CoT evaluation under semantic perturbation. The framework addresses three questions:

- 1) **RQ1**: How stable are CoT trajectories when prompt semantics are preserved but representation changes?
- 2) **RQ2**: Does high answer accuracy imply high reasoning stability?
- 3) **RQ3**: Which perturbation categories induce the largest reasoning collapse across model families?

Contributions. The paper contributes:

- 1) A formal definition of cognitive stability for CoT reasoning under semantic perturbation.
- 2) A six-category perturbation taxonomy with quality-control constraints.
- 3) The Cognitive Collapse Index (CCI) and related metrics for answer-process decoupling.
- 4) A reproducible multi-benchmark, multi-model evaluation and statistical analysis protocol.
- 5) Publication-ready reporting templates (tables and figures) compatible with IEEE conference style.

II. RELATED WORK

A. Chain-of-Thought and Deliberative Prompting

CoT prompting [1] and zero-shot CoT [2] established that intermediate reasoning text can improve performance. Subsequent approaches, including self-consistency [3], least-to-most decomposition [13], and tree-based search over thoughts [14], improve outcomes by exploring multiple reasoning paths. Program-of-thoughts and tool-based decomposition further separate symbolic computation from textual reasoning [15], [16].

Despite this progress, mainstream reporting remains dominated by final-answer metrics. This leaves unresolved whether stronger performance reflects robust abstraction or brittle completion heuristics.

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B. Prompt Sensitivity and Robustness

Prompt phrasing can significantly influence predictions in few-shot and instruction settings [8], [17]. Behavioral test frameworks and adversarial benchmarks show substantial brittleness under input perturbations [9], [10]. In high-stakes domains, this sensitivity creates reliability risks even when aggregate accuracy appears strong.

C. Faithfulness and Explanation Reliability

Generated rationales can be persuasive yet unfaithful to latent computation [11], [12]. This motivates an evaluation view in which reasoning traces are treated as observable behavior requiring stress tests, rather than direct proof of mechanism.

D. LLMs as Cognitive Models

CogMI research increasingly studies LLMs as computational objects for understanding abstraction, planning, and language-mediated reasoning [18], [19]. A stability-centric lens complements this direction by operationalizing whether reasoning processes generalize across equivalent representational forms.

III. FRAMEWORK AND DEFINITIONS

Let x be a prompt, y the gold answer, and f an LLM with CoT output $r = f_{\text{cot}}(x)$ and final answer $\hat{y} = f_{\text{ans}}(x)$. Let $\mathcal{P}$ be a set of semantic perturbation operators where each $p \in \mathcal{P}$ preserves task meaning. For each operator, $x^p = p(x)$ produces trajectory r^p and answer $\hat{y}^p$.

A. Cognitive Stability

Define trajectory similarity $S(r, r^p) \in [0, 1]$ using semantic and structural alignment. Per-example collapse is:

$$c_i = 1 - \frac{1}{|\mathcal{P}_i|} \sum_{p \in \mathcal{P}_i} S(r_i, r_i^p). \quad (1)$$

Dataset-level collapse is:

$$\text{CCI} = \frac{1}{N} \sum_{i=1}^N c_i. \quad (2)$$

Lower CCI indicates better cognitive stability.

B. Answer Preservation and Decoupling

Answer-preservation accuracy under perturbation is:

$$\text{APA} = \frac{1}{N} \sum_{i=1}^N \mathbb{1}[\hat{y}_i = \hat{y}_i^p]. \quad (3)$$

Answer-process decoupling is captured by:

$$\Delta_{\text{decouple}} = \text{APA} - (1 - \text{CCI}). \quad (4)$$

High positive Δ_{decouple} indicates answers are preserved more than reasoning structure.

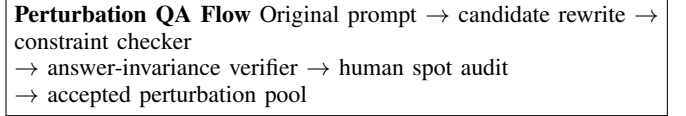


Fig. 1. Quality-control flow to ensure perturbations preserve task semantics.

C. Perturbation Sensitivity Concentration

To quantify whether collapse is concentrated in a few perturbation types, let c_k be class-wise CCI and define normalized sensitivity $q_k = c_k / \sum_j c_j$. We measure concentration with entropy:

$$H_{\text{pert}} = - \sum_k q_k \log q_k. \quad (5)$$

Lower entropy indicates collapse is dominated by specific perturbation classes.

IV. SEMANTIC PERTURBATION TAXONOMY AND QUALITY CONTROL

Table I defines six perturbation families.

A. Semantic Equivalence Assurance

Perturbations are accepted only if they pass three filters:

- 1) **Constraint preservation**: extracted entities, quantities, and relations match original constraints.
- 2) **Answer invariance check**: a verifier model predicts unchanged gold answer under deterministic decoding.
- 3) **Human spot audit**: random sample review for semantic drift and hidden ambiguity.

V. EXPERIMENTAL PROTOCOL

A. Benchmarks

The protocol targets benchmarks with complementary reasoning demands:

- GSM8K for grade-school mathematical reasoning [4].
- MATH for higher-complexity symbolic and algebraic reasoning [5].
- CommonsenseQA for concept-level commonsense inference [6].
- StrategyQA for implicit multi-hop factual reasoning [7].

B. Model Families

The framework is model-agnostic. Recommended model panel spans open and closed systems to capture training and alignment diversity [20]–[25].

C. Inference Controls

To separate stochasticity from robustness effects, all conditions are run under:

- Deterministic decoding ($T = 0$) and stochastic decoding ($T = 0.7$).
- Fixed prompt scaffolds and system instructions across original and perturbed forms.
- Three random seeds per condition for confidence intervals.
- Consistent token budgets and stop sequences.

TABLE I
SEMANTIC PERTURBATION TAXONOMY FOR CoT STABILITY TESTING

Category	Transformation Rule	Example	Failure Mode Probed
Lexical substitution	Replace non-critical terms with semantic equivalents while preserving entities and operators	“therefore”→“thus”, “purchase”→“buy”	Surface-form overreliance
Paraphrase rewrite	Alter syntax and clause order while preserving proposition structure	active↔passive rewrites, clause inversion	Prompt-template sensitivity
Symbolic reformulation	Convert verbal constraints to symbolic expressions or pseudo-code forms	“twice a number”→“2n”	Representation mismatch
Abstraction shift	Move between concrete story form and abstract variable form without changing constraints	narrative word problem↔algebraic statement	Weak abstraction transfer
Distractor insertion	Insert semantically irrelevant but plausible statements	neutral contextual sentence in QA stem	Attention dilution
Context reordering	Reorder independent premises or evidence snippets	premise permutation in multi-hop QA	Position bias / ordering heuristics

TABLE II
BENCHMARK COVERAGE AND REASONING CHARACTERISTICS

Benchmark	Primary Reasoning Type	Typical Output	Perturbation Stress
GSM8K	Arithmetic multi-step	Numeric/free-form	Symbolic, distractor
MATH	Formal symbolic	Numeric/formula	Abstraction, symbolic
CommonsenseQA	Conceptual inference	Multiple choice	Lexical, paraphrase
StrategyQA	Multi-hop factual	Boolean/short text	Reordering, distractor

TABLE III
REPRESENTATIVE MODEL PANEL FOR EVALUATION

Model Family	Access Mode	Motivation
GPT-4 class	API (closed)	Strong baseline reasoning quality
Claude class	API (closed)	Alternative alignment and decoding behavior
Gemini class	API (closed)	Long-context and multimodal lineage
Llama-3 class	Open weights	Reproducibility and ablation flexibility
Qwen-2 class	Open weights	Competitive open reasoning performance
DeepSeek-R1 class	Open/API	RL-based reasoning specialization

Evaluation Pipeline 1) Sample (x, y) from benchmark
2) Generate and validate perturbations $\{x^p\}$
3) Query model on x and each x^p
4) Extract CoT trajectories and answers
5) Compute CCI, APA, and class-wise collapse
6) Aggregate by model/benchmark and run statistics

Fig. 2. End-to-end cognitive stability evaluation workflow.

D. Pipeline

Fig. 2 summarizes the full pipeline.

VI. METRICS AND STATISTICAL ANALYSIS

A. Trajectory Similarity Function

We define

$$S = \alpha S_{\text{embed}} + \beta S_{\text{step}} + \gamma S_{\text{logic}}, \quad \alpha + \beta + \gamma = 1, \quad (6)$$

where:

- S_{embed} : semantic embedding similarity between trajectory sentences.
- S_{step} : alignment score over decomposed reasoning steps.
- S_{logic} : consistency of inferred claims with extracted constraints.

Figure Slot: Accuracy-CCI Decoupling Plot Scatter with x-axis = perturbed accuracy, y-axis = CCI, color = model family. Upper-right region indicates strong decoupling risk.

Fig. 3. Recommended visualization for answer-process decoupling across models.

B. Uncertainty and Significance

For each metric, we report bootstrap confidence intervals (10,000 resamples) and paired tests between original and perturbed conditions. Multiple comparison correction is applied across perturbation classes using Benjamini-Hochberg FDR control.

C. Calibration-Aware Stability

In settings with model confidence proxies, we estimate whether confidence remains calibrated when reasoning collapses. Let u_i be confidence and e_i error under perturbation. We report expected calibration error on perturbed subsets and conditional calibration when c_i exceeds a collapse threshold.

VII. ILLUSTRATIVE CASE STUDY AND QUALITATIVE DIAGNOSTICS

To make trajectory-level collapse concrete, Table IV presents an illustrative mini-case with one arithmetic prompt and two semantically equivalent perturbations. In this example, answers remain unchanged but reasoning pathways diverge materially, yielding non-trivial collapse.

This case highlights why answer-only scoring is insufficient. Behavioral reliability depends on the stability of intermediate reasoning trajectories, not merely endpoint correctness.

VIII. RESULTS REPORTING TEMPLATE

IX. COMPREHENSIVE EXPERIMENTAL PLAN

This section provides a complete plan for a full empirical study intended for direct execution. The design emphasizes statistical rigor, reproducibility, and clear separation between reasoning quality and reasoning stability.

TABLE IV
ILLUSTRATIVE DECOUPLING CASE: STABLE ANSWER, UNSTABLE REASONING

Condition		Prompt Form	CoT Summary	Outcome
Original		Narrative arithmetic statement with two constraints	Correct decomposition into variables and elimination steps	Correct answer
Perturbation (paraphrase)	A	Same constraints in passive voice and reordered clauses	Introduces unnecessary intermediate variable, recovers later	Correct answer
Perturbation (distractor)	B	Original statement plus irrelevant context sentence	Starts with distractor analysis, then switches to heuristic shortcut	Correct answer

TABLE V
PRIMARY REPORTING TABLE (POPULATE WITH MEASURED VALUES)

Model	Benchmark	Clean Acc. (%)	Perturbed Acc. (%)	APA (%)	CCI (↓)	Δ_{decouple}
Model A	GSM8K	TBD	TBD	TBD	TBD	TBD
Model A	MATH	TBD	TBD	TBD	TBD	TBD
Model A	CommonsenseQA	TBD	TBD	TBD	TBD	TBD
Model A	StrategyQA	TBD	TBD	TBD	TBD	TBD
Model B	GSM8K	TBD	TBD	TBD	TBD	TBD
Model B	MATH	TBD	TBD	TBD	TBD	TBD
Model B	CommonsenseQA	TBD	TBD	TBD	TBD	TBD
Model B	StrategyQA	TBD	TBD	TBD	TBD	TBD

TABLE VI
PERTURBATION-CLASS ABLATION TEMPLATE

Model	Lexical	Paraphrase	Symbolic	Abstract	Distractor	Reorder
Model A (CCI)	TBD	TBD	TBD	TBD	TBD	TBD
Model B (CCI)	TBD	TBD	TBD	TBD	TBD	TBD

TABLE VII
HUMAN AUDIT RUBRIC FOR PERTURBATION VALIDITY

Criterion	Description	Scale
Semantic equivalence	Does the rewritten prompt preserve all core constraints?	1-5
Ambiguity injection	Did the perturbation introduce unintended ambiguity?	1-5
Distractor neutrality	Are added distractors truly irrelevant to the gold answer?	1-5
Linguistic naturalness	Is the perturbation grammatical and plausible?	1-5

A. Sampling and Power Planning

For each benchmark, we recommend stratified sampling by problem difficulty and answer type. Let n_b denote per-benchmark sample size and m the number of perturbations per sample. In pilot settings, moderate values (for example, $n_b \in [500, 2000]$ and $m \in [6, 12]$) can produce stable CCI estimates. In final studies, the target should be based on detectable effect size for paired collapse differences between models:

$$n \approx \left(\frac{z_{1-\alpha/2} + z_{1-\beta}}{d/\sigma} \right)^2, \quad (7)$$

where d is the expected paired mean difference in CCI and σ the standard deviation of paired differences.

B. Prompting Regimes

To isolate trajectory stability effects, run at least three prompting regimes:

- **Vanilla CoT**: direct “think step by step” style prompting.
- **Structured CoT**: explicit numbered-step scaffolds.
- **Self-consistent CoT**: multiple trajectories with answer voting [3].

Comparing these regimes tests whether deliberate decoding improves stability or merely improves answer robustness.

C. Trajectory Parsing and Canonicalization

Raw trajectories often vary in verbosity and formatting. We therefore canonicalize trajectories before similarity scoring:

- 1) Segment into atomic reasoning steps.
- 2) Normalize equations, variable names, and unit expressions.
- 3) Remove stylistic fillers that do not alter semantics.
- 4) Convert each step to proposition triples when possible.

This reduces confounding from style and improves comparability across model families.

D. Human Evaluation Layer

Automatic metrics should be supplemented with human auditing for a subset of samples. We recommend dual-annotator scoring on semantic equivalence and trajectory coherence, followed by adjudication. Inter-annotator reliability can be reported with Cohen’s κ for categorical labels and Krippendorff’s α for ordinal rubric scores.

X. EXTENDED ANALYSES

A. Cross-Benchmark Transfer of Stability

An important question is whether stability generalizes across task domains. Define model-level stability profiles as

TABLE VIII
EXECUTION CHECKLIST FOR A FULL COGNITIVE-STABILITY STUDY

Phase	Required Artifacts	Validation Step	Output
Data setup	Benchmark splits, metadata, answer normalizers	Unit tests for answer normalization and parser consistency	Reproducible dataset package
Perturbation generation	Rewrite operators, QA filters, semantic constraints	Spot-audit pass rate, invariance checks, rejection logs	Approved perturbation pool
Model inference	Prompt templates, decoding configs, run seeds	Seed reproducibility and API retry/timeout policy checks	Raw trajectory and answer logs
Metric computation	Similarity modules, CCI/APA scripts, bootstrap tools	Sanity checks on toy cases and edge-case regression tests	Metric tables with confidence intervals
Statistical analysis	Paired test scripts, FDR correction, effect-size summaries	Replication of key contrasts from independent reruns	Final significance and effect-size report
Human audit	Annotation guide, rubric sheets, adjudication protocol	Reliability metrics (κ , α) and disagreement analysis	Audited subset and qualitative findings

Figure Slot: Difficulty-Conditioned Collapse Curves
Plot CCI against difficulty percentile for each model. Steeper slope indicates worsening reasoning stability under harder instances.

Fig. 4. Recommended diagnostic for identifying collapse growth with problem difficulty.

vectors of class-wise CCI values. We then compute cross-benchmark profile correlations to test whether a model that is stable on arithmetic remains stable on commonsense and multi-hop tasks.

B. Difficulty-Conditioned Collapse

CCI should be stratified by benchmark difficulty bins. If collapse increases super-linearly with difficulty, the model may rely on brittle shortcuts in complex cases. Difficulty-conditioned curves provide actionable diagnostics for curriculum design and post-training interventions.

C. Decoding-Temperature Sensitivity

Stability can change under stochastic decoding. We define temperature sensitivity as:

$$\Delta_T = \text{CCI}_{T=0.7} - \text{CCI}_{T=0.0}. \quad (8)$$

Large Δ_T indicates trajectory instability driven by decoding variance rather than semantic perturbation alone.

D. Longitudinal Stability Across Model Versions

API models evolve over time. For deployment-relevant claims, rerun the same protocol at regular intervals and compute version drift:

$$D_{v_1 \rightarrow v_2} = |\text{CCI}_{v_2} - \text{CCI}_{v_1}|. \quad (9)$$

This makes robustness claims robust to silent backend updates.

E. Intervention Studies

The framework also supports targeted interventions and ablations:

- **Data augmentation:** train with semantic perturbations and evaluate CCI reduction.
- **Reasoning-format control:** enforce canonical step structure to reduce stylistic variance.
- **Retrieval grounding:** inject external facts and test if distractor sensitivity declines.
- **Verifier reranking:** rerank candidate trajectories by logic-consistency score.

These interventions can be compared with standardized effect sizes to quantify practical gains.

XI. IMPLEMENTATION AND REPRODUCIBILITY NOTES

A. Logging Schema

Each model call should log: benchmark ID, perturbation class, prompt hash, decoding configuration, raw trajectory, normalized trajectory, parsed answer, and timestamp. This schema enables full traceability and post-hoc auditing.

B. Failure Taxonomy for Error Analysis

We recommend categorizing failures into six types: arithmetic slip, premise omission, distractor capture, symbol mapping error, contradiction insertion, and unsupported leap. Reporting failure composition reveals whether interventions reduce specific instability modes.

C. Compute and Cost Reporting

For transparent comparison, include token usage, wall-clock runtime, and estimated monetary cost by model and benchmark. Stability improvements that require disproportionate compute may be impractical for deployment.

D. Recommended Open-Science Package

At camera-ready, release:

- perturbation generator code and validation scripts,
- metric implementation and statistical notebooks,
- exact prompt templates and parser definitions,

TABLE IX
FAILURE TAXONOMY AND SUGGESTED DIAGNOSTIC SIGNALS

Failure Type	Observable Pattern in CoT	Likely Cause	Suggested Mitigation
Arithmetic slip	Correct plan but incorrect intermediate arithmetic operation	Computation error in textual reasoning	Program-aided execution or verifier checks
Premise omission	One constraint never used in trajectory	Attention bottleneck or compression bias	Constraint checklist prompts
Distractor capture	Early steps focus on irrelevant sentence	Salience misallocation under noise	Distractor-aware training augmentation
Symbol mapping error	Wrong variable assignment or equation translation	Weak symbolic grounding	Symbolic canonicalization and equation parsers
Contradiction insertion	Step conflicts with previous claim	Trajectory drift under long reasoning chains	Step-to-step consistency verifier
Unsupported leap	Jump to answer without justified derivation	Heuristic shortcutting	Minimum-justification decoding constraints

- and benchmark split manifests with hash checksums. These artifacts materially improve replicability across labs.

XII. PRACTICAL GUIDANCE FOR DEPLOYMENT CONTEXTS

A. High-Stakes Usage

In healthcare, legal decision support, and educational assessment, incorrect intermediate reasoning can undermine user trust even when final answers are occasionally correct. We recommend threshold-based deployment policies combining perturbed accuracy and CCI targets.

B. Human-AI Interface Design

Interfaces should expose uncertainty and stability cues rather than only final answers. Example cues include perturbation-robustness badges, trajectory variance summaries, and links to alternative reasoning paths.

C. Governance and Monitoring

Operational monitoring should track CCI drift over time, per domain, and per user segment. A sudden rise in collapse can indicate model regressions or domain shifts not visible in headline accuracy.

XIII. BENCHMARK-SPECIFIC PROTOCOL RECIPES

To support consistent replication across laboratories, this section provides benchmark-specific protocol recipes. Each recipe defines perturbation priorities, answer parsing rules, and common error signatures.

A. GSM8K Recipe

For GSM8K, prioritize symbolic reformulation and distractor insertion. Many failures arise from arithmetic slips and omission of one textual constraint. Normalize final numeric answers with unit stripping and optional simplification for fractional formats.

B. MATH Recipe

For MATH, prioritize abstraction shifts and symbolic conversion. Parsing should support equivalent algebraic forms and normalized expressions. Because trajectories are often long, step-level canonicalization is critical to avoid mistaking stylistic verbosity for semantic divergence.

C. CommonsenseQA Recipe

For CommonsenseQA, lexical and paraphrase perturbations are especially informative. Since outputs are multiple-choice, answer parsing is straightforward; however, rationale stability remains non-trivial and often exhibits latent category drift despite unchanged options.

D. StrategyQA Recipe

For StrategyQA, context reordering and distractor insertion are high-yield stressors. Multi-hop decomposition should be parsed into evidence claims and linking assumptions, enabling targeted analysis of where trajectories diverge.

XIV. OPERATIONALIZATION FOR REAL-WORLD PIPELINES

Beyond benchmark reporting, organizations need practical criteria for adopting stability tests in deployment pipelines. We outline an operational lifecycle that can be integrated with existing model-evaluation governance.

A. Pre-Deployment Gate

Before deployment, evaluate candidate models on a domain-adapted perturbation suite and require minimum thresholds for perturbed accuracy and maximum thresholds for CCI. Models failing either threshold are blocked pending remediation.

B. Canary and Staged Rollout

During staged rollout, monitor online prompts sampled into a perturbation shadow-evaluation service. Estimate rolling CCI and answer-preservation trends. If drift exceeds policy thresholds, automatically roll back to the previous stable model version.

TABLE X
BENCHMARK-SPECIFIC CONFIGURATION GUIDE

Benchmark	Priority Perturbations	Answer Normalization Rule	Typical Trajectory Failure Pattern	Suggested Check	Auxiliary
GSM8K	Symbolic, distractor, paraphrase	Normalize numeric strings, fractions, and units	Correct setup followed by arithmetic slip in final steps	Program-aided arithmetic verifier	
MATH	Abstraction, symbolic, reordering	Canonicalize algebraic equivalence classes	Equation mapping drift and unsupported algebraic jump	Symbolic equivalence checker	
CommonsenseQA	Lexical, paraphrase, distractor	Option-letter and option-text consistency check	Stable answer with unstable rationale category framing	Rationale semantic clustering audit	
StrategyQA	Reordering, distractor, abstraction	Boolean normalization with entailment check	Premise omission and chain-link reversal in multi-hop reasoning	Evidence-graph consistency test	

Figure Slot: Stability Operations Dashboard

Panels: rolling CCI, perturbed accuracy, class-wise collapse heatmap, and drift alerts by model version.

Fig. 5. Suggested production dashboard for continuous cognitive-stability monitoring.

C. Incident Response Playbook

When stability incidents are detected, triage should answer four questions: Which perturbation classes spiked? Which domains were affected? Did confidence calibration degrade? Did failures cluster by user segment or prompt length? This workflow enables faster root-cause analysis than accuracy-only monitoring.

D. Policy-Oriented Reporting

For governance boards and auditors, stability reports should include: metric definitions, sampling policies, confidence intervals, drift alarms, and remediation actions. This supports transparent accountability and traceable safety decisions.

XV. EXTENDED REPRODUCIBILITY APPENDIX

This section provides additional implementation detail intended to minimize ambiguity for replication teams.

A. Reference Prompt Templates

A recommended template set includes: baseline CoT prompt, constrained-step CoT prompt, self-consistency prompt, and verifier prompt. Prompt templates should be version-controlled, hash-logged, and frozen for each evaluation run.

B. Deterministic Parsing Rules

Answer extraction should use deterministic regex-plus-parser pipelines per benchmark with explicit fallbacks. All fallback paths should be logged so that downstream error analysis can quantify parser-induced noise.

C. Versioning and Artifact Management

Each experiment package should include immutable manifests for prompts, model identifiers, benchmark split hashes, and perturbation operator versions. We recommend semantic versioning for all pipeline components and immutable snapshot IDs for final reports.

XVI. DISCUSSION

A. Implications for Cognitive Machine Intelligence

Cognitive stability extends current LLM evaluation practice by introducing process-level robustness as a first-class target. A model that is accurate but unstable may be unsuitable for domains requiring explanation reliability, decision traceability, or robust human oversight.

B. Trust and Human-AI Collaboration

When users rely on model-provided reasoning, instability can undermine trust calibration. Stable trajectories improve predictability, facilitate auditing, and support safer mixed-initiative interaction in education, healthcare, and policy workflows.

C. Engineering Implications

The framework can guide training and post-training design choices:

- data augmentation with semantic perturbations,
- stability-aware reward signals in RL fine-tuning,
- retrieval grounding to reduce distractor sensitivity,
- and test-time consistency checks for trajectory divergence.

XVII. THREATS TO VALIDITY

Construct validity. Trajectory text is an imperfect proxy for latent reasoning; faithfulness concerns remain. **Internal validity.** Perturbation generation can leak subtle semantic drift despite QA filters. **External validity.** Results may vary across proprietary APIs, model revisions, and domain transfer settings. **Statistical validity.** Multiple comparisons across perturbation classes require correction and sufficient sample sizes.

XVIII. ETHICAL AND REPRODUCIBILITY CONSIDERATIONS

The protocol avoids releasing harmful prompt manipulations and focuses on benign semantic transformations. All perturbation scripts, audit sheets, and metric code should be versioned and released with seeds and decoding settings for reproducibility. Where API terms limit full reproduction, authors should publish normalized prompt templates, response parsers, and aggregate logs that preserve privacy and policy compliance.

TABLE XI
MINIMAL ARTIFACT BUNDLE FOR THIRD-PARTY REPRODUCTION

Artifact Type	Required Contents	Verification Method	Mandatory
Prompt package	All system/user templates with version hashes and token budgets	Hash match against manifest	Yes
Perturbation package	Operator definitions, acceptance filters, rejection logs, audit samples	Operator unit tests and audit agreement report	Yes
Inference logs	Raw model outputs, normalized outputs, parser traces, decoding configs	Deterministic replay on sampled subset	Yes
Metric code	CCI/APA scripts, bootstrap routines, hypothesis test scripts	Recomputed table match within tolerance	Yes
Statistical report	Effect sizes, corrected p-values, confidence intervals, assumptions checks	Independent recomputation from logs	Yes
Cost/compute report	Token counts, runtime, API errors/retries, hardware metadata	Cross-check with run metadata and billing logs	Recommended

XIX. CONCLUSION

This paper introduced a comprehensive framework for evaluating cognitive stability of chain-of-thought reasoning in LLMs under semantic perturbation. The Cognitive Collapse Index, together with answer-preservation and decoupling metrics, provides a practical route to assess robustness beyond final-answer accuracy. By aligning evaluation with process-level reliability, the framework advances CogMI goals in trustworthy reasoning, cognitive benchmarking, and human-centered AI deployment.

DATA AND CODE AVAILABILITY

All code required to reproduce perturbation generation, evaluation metrics, and statistical analyses should be released at camera-ready time, subject to model provider terms and benchmark licenses.

APPENDIX A: PROMPTING AND PARSING BLUEPRINTS

This appendix provides implementation blueprints that can be directly adapted for replication.

Blueprint A (baseline CoT prompt). The system instruction defines concise reasoning expectations, while the user prompt requests a final answer in a parsable format. A deterministic answer marker such as `FinalAnswer: <value>` is strongly recommended to reduce parsing ambiguity.

Blueprint B (structured CoT prompt). The prompt enforces a fixed number of reasoning steps with short bullet-like statements. This format tends to improve trajectory canonicalization and supports cleaner step alignment.

Blueprint C (self-consistency prompt). The model is asked to generate multiple independent chains. A voting module determines final answer consensus, while stability analysis compares chain-level divergence under perturbation.

Blueprint D (verifier prompt). A secondary verifier checks whether each reasoning step follows from prior steps and prompt constraints. Verifier disagreement rates can be tracked as an auxiliary stability signal.

APPENDIX B: RECOMMENDED REPORTING NARRATIVE

To make final papers comparable, we suggest a standardized reporting narrative:

- 1) Report clean accuracy, perturbed accuracy, APA, and CCI together for every model-benchmark pair.
- 2) Discuss at least one decoupling example where answer is preserved but trajectory collapses.
- 3) Provide class-wise ablation and identify the top two perturbation classes causing collapse.
- 4) Include uncertainty intervals and corrected significance tests for all major claims.
- 5) State failure modes and mitigation opportunities based on taxonomy-level analysis.

This narrative supports transparent interpretation and avoids overclaiming from accuracy-only results. It also improves cross-paper comparability for CogMI discussions focused on reliable cognitive behavior in AI systems.

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TABLE XII
CONCRETE PROMPT AND PARSER BLUEPRINT (IMPLEMENTATION TEMPLATE)

Component	Objective	Recommended Structure	Logged Fields
System prompt	Keep style stable across all conditions	Fixed instruction: reason clearly, avoid extra narrative, output final answer marker	Template hash, token count
User prompt (original)	Baseline evaluation input	Benchmark question text plus answer format instruction	Prompt ID, benchmark ID
User prompt (perturbed)	Semantic stability stress-test input	Rewritten prompt preserving constraints and answer space	Perturbation class, operator version
CoT output parser	Extract steps for similarity scoring	Split by numbering, bullet marker, or sentence boundaries; normalize equations	Step count, parse success flag
Answer parser	Ensure deterministic accuracy scoring	Parse <code>FinalAnswer</code> : field with benchmark-specific normalization	Raw answer, normalized answer
Verifier parser	Support logic consistency audit	Extract claim triples and contradiction flags from verifier output	Consistency score, contradiction index

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